

# Akamai SIEM Integration with Splunk Cloud (Victoria)

Onboarding, CIM Validation, and Detection Readiness

## Fact-Checked Reference

|                 |                                                 |
|-----------------|-------------------------------------------------|
| Classification  | Internal -- Security Engineering                |
| Platform        | Splunk Cloud Victoria Experience (Dev + Prod)   |
| ES Version      | Enterprise Security (current supported release) |
| RBA Status      | Not enabled                                     |
| Akamai Products | WAF (App & API Protector), DDoS, Bot Manager    |
| Date            | February 03, 2026                               |
| Version         | 1.0                                             |

**Fact-Checking Standard:** Every technical claim in this document has been validated against official Akamai TechDocs ([techdocs.akamai.com/siem-integration](https://techdocs.akamai.com/siem-integration)) and official Splunk documentation ([docs.splunk.com](https://docs.splunk.com)). Where a behavior is not explicitly documented, this document states: *"This behavior depends on deployment and must be validated in the environment."*

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## 1. Executive Summary

This document provides a fact-checked, delivery-ready reference for integrating Akamai security events into Splunk Cloud (Victoria Experience) for consumption by Splunk Enterprise Security (ES). It is intended for security engineering managers and senior engineers responsible for onboarding Akamai WAF, DDoS, and Bot Manager data into a production SIEM environment.

Akamai operates a global Content Delivery Network (CDN) and edge security platform. When Akamai security products -- such as WAF (App & API Protector), DDoS protection, and Bot Manager -- detect threats at the edge, they generate security events. These events are stored temporarily (for 12 hours) in the Akamai Security Events Collector (ASEC) and are exposed via the **Akamai SIEM Integration API**.

Critically, Splunk Cloud does not poll the Akamai API directly. An external **SIEM connector** must run on customer-managed infrastructure to retrieve events and forward them to Splunk Cloud via HTTP Event Collector (HEC). This architectural boundary is non-negotiable in Splunk Cloud deployments.

### What This Document Covers

- The end-to-end architecture from Akamai edge to Splunk ES
- What Akamai officially provides versus what the engineering team must implement
- CIM (Common Information Model) mapping requirements and validation procedures
- A Dev-to-Prod safety model to prevent breaking Enterprise Security in production
- Detection readiness criteria and the distinction between data readiness and active detections

**Key Principle:** This integration is not plug-and-play. Akamai provides the API and a baseline Splunk add-on with partial CIM coverage. The engineering team owns the collector, the CIM validation, the CIM extension, and the promotion from Dev to Prod.

## 2. Akamai Security Products Overview

Akamai's edge security platform includes several products that generate security events relevant to SIEM integration. Each product operates at the Akamai edge and produces events that flow through the same SIEM Integration API, but each has distinct event characteristics.

| Product                                     | Description                                                                                                                                                                                                                                                                                                            | Key Event Fields                                                                                                                                                   |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Akamai WAF (App &amp; API Protector)</b> | Inspects HTTP/HTTPS requests at the edge against rulesets (Kona Rule Set, Adaptive Security Engine). Detects SQL injection, XSS, command injection, file inclusion, and protocol violations. Events include rule IDs, rule messages, rule actions (alert, deny, monitor), and the full HTTP request context.           | attackData.rules, attackData.ruleActions, attackData.ruleMessages, attackData.ruleTags, httpMessage.method, httpMessage.host, httpMessage.path, httpMessage.status |
| <b>Akamai DDoS Protection</b>               | Detects and mitigates volumetric and application-layer DDoS attacks at the edge. Events capture rate-control triggers, slow POST detections, and IP-based deny actions. DDoS events share the same JSON structure as WAF events.                                                                                       | attackData.ruleActions (deny/alert), attackData.rules (rate control rule IDs), httpMessage.status (403 for blocked requests)                                       |
| <b>Akamai Bot Manager</b>                   | Classifies incoming requests as human, known bot, or unknown bot. Assigns a bot score and categorizes by bot type. Events include bot classification data, bot score, and response segment. Bot Manager events are included in SIEM output when 'Include Bot Manager Events' is enabled in the security configuration. | botData.botScore, botData.responseSegment, attackData.clientIP, attackData.configId, httpMessage.requestHeaders.User-Agent                                         |

### How Events Are Exposed

All security events from the products above are exposed through a single **Akamai SIEM Integration API** endpoint. Per Akamai documentation, the API endpoint is:

```
GET /siem/v1/configs/{configId}
```

Each Akamai security configuration has a unique **configId**. An enterprise may have multiple config IDs -- one per security product, per web property, or per business unit. The API accepts multiple config IDs separated by semicolons:

```
GET /siem/v1/configs/{configId1};{configId2};{configId3}
```

Events are returned as a series of JSON objects, one per line. Each object contains three primary sections: **attackData** (rule and attacker information), **httpMessage** (the HTTP request/response context), and **geo** (geolocation of the client IP). The last line of the response contains offset context for incremental polling.

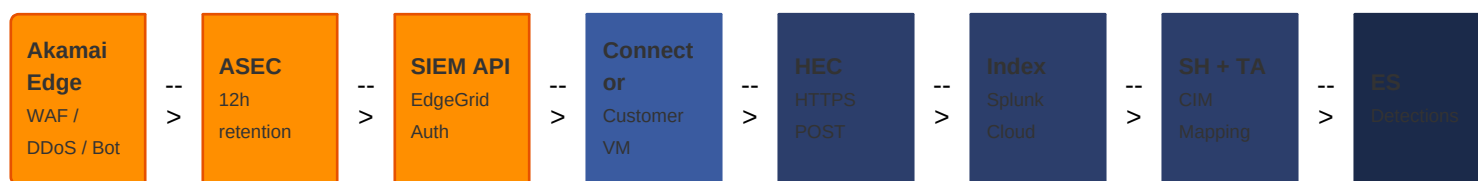
Source: Akamai TechDocs -- SIEM Integration API Reference ([techdocs.akamai.com/siem-integration/reference/api](https://techdocs.akamai.com/siem-integration/reference/api))

### 3. Akamai to Splunk Cloud Architecture (Fact-Checked)

The integration follows a linear, unidirectional data flow. There are no alternative architectures for Splunk Cloud Victoria Experience. The following describes each stage of the pipeline.

| Stage | Component                                       | Description                                                                                                                                                                                                                            |
|-------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Akamai Edge                                     | Akamai CDN processes web traffic. Security products (WAF, DDoS, Bot Manager) evaluate each request against configured policies. Requests that trigger a rule generate a security event.                                                |
| 2     | Akamai Security Events Collector (ASEC)         | Events are stored in the ASEC for 12 hours. Per Akamai documentation, events older than 12 hours cannot be retrieved. The ASEC is Akamai-managed infrastructure; the customer has no direct access.                                    |
| 3     | SIEM Integration API                            | The REST API (GET /siem/v1/configs/{configId}) exposes events from the ASEC. Supports offset-based incremental polling and time-range queries. Requires EdgeGrid authentication (client_token, client_secret, access_token, hostname). |
| 4     | Akamai SIEM Connector (Customer Infrastructure) | An external process that runs on a customer-managed Linux VM, Docker container, or serverless function. It polls the API, manages offsets, and forwards events. This is the ONLY component that communicates with the Akamai API.      |
| 5     | Splunk Cloud HEC                                | The connector POSTs JSON events to the Splunk Cloud HTTP Event Collector endpoint. HEC assigns the configured index, sourcetype, and source at ingest time.                                                                            |
| 6     | Splunk Cloud Index                              | Events are indexed and stored. The index is created via the Splunk Cloud UI. Retention and access controls are configured per organizational policy.                                                                                   |
| 7     | Search Head + TA                                | The Akamai SIEM Integration add-on (Splunkbase App 4310) is installed on the search head. At search time, it provides JSON field extraction, FIELDALIAS rules, EVAL transformations, and CIM tag/eventtype definitions.                |
| 8     | Enterprise Security                             | ES consumes CIM-normalized data from accelerated data models (primarily Web, optionally Intrusion_Detection). Correlation searches, dashboards, and threat intelligence lookups operate on CIM field names.                            |

#### Data Flow Diagram



**Architectural Boundary:** The SIEM connector on customer infrastructure is the ONLY component that communicates with the Akamai API. Splunk Cloud receives data passively via HEC. Akamai EdgeGrid API credentials are stored ONLY on the connector host -- never in Splunk Cloud.

## 4. Akamai SIEM Connector (Collector)

### What Akamai Officially Provides

Akamai provides official SIEM connector packages designed to run on customer infrastructure. Per Akamai documentation, the connector is installed behind the corporate firewall and makes periodic calls to the SIEM API to collect JSON event data in near real time.

| Attribute       | Detail                                                                                                                                                                                                                      |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Official Name   | Akamai SIEM Connector                                                                                                                                                                                                       |
| Available From  | Akamai TechDocs ( <a href="https://techdocs.akamai.com/siem-integration">techdocs.akamai.com/siem-integration</a> ), GitHub ( <a href="https://github.com/akamai/ULS">github.com/akamai/ULS</a> for Universal Log Streamer) |
| Connector Types | CEF Syslog connector, Splunk-specific connector (Java-based modular input in TA), Universal Log Streamer (ULS)                                                                                                              |
| Runs On         | Customer-managed infrastructure: Linux VM, Docker container, serverless function, or customer Heavy Forwarder                                                                                                               |
| Authentication  | Akamai EdgeGrid: client_token, client_secret, access_token, hostname (provisioned in Akamai Control Center)                                                                                                                 |
| Configuration   | Config ID(s), polling interval, offset management, destination HEC endpoint URL + token                                                                                                                                     |
| Data Retention  | ASEC retains events for 12 hours. If the connector is offline for more than 12 hours, events are permanently lost.                                                                                                          |

### Why Splunk Cloud Cannot Run This Connector

In Splunk Cloud (Victoria Experience), the customer does not have OS-level access to the search heads or indexers. The following constraints apply:

- No ability to install arbitrary software or run external processes on Splunk Cloud infrastructure
- No ability to execute scheduled scripts, cron jobs, or persistent background processes
- Java-based modular inputs (included in the Akamai TA) have compatibility constraints and are not universally reliable on Victoria Experience
- Storing Akamai EdgeGrid credentials inside Splunk Cloud would violate the principle that API credentials belong only on customer-managed infrastructure

### What the Connector Does

| Responsibility | Detail                                                           |
|----------------|------------------------------------------------------------------|
| Authenticates  | Presents EdgeGrid credentials to the Akamai SIEM API             |
| Polls          | Calls GET /siem/v1/configs/{configId} on a configurable interval |

| Responsibility  | Detail                                                                                       |
|-----------------|----------------------------------------------------------------------------------------------|
| Manages Offsets | Stores the offset from the API ResponseContext to enable incremental, deduplicated retrieval |
| Forwards        | POSTs JSON events to the Splunk Cloud HEC endpoint via HTTPS                                 |

## What Remains Our Responsibility

| Responsibility         | Detail                                                         |
|------------------------|----------------------------------------------------------------|
| Collector availability | Monitoring, restarts, patching the host OS                     |
| Credential rotation    | Akamai EdgeGrid credentials + Splunk HEC tokens                |
| Config ID management   | Adding new config IDs as Akamai security configurations change |
| HEC endpoint health    | Monitoring for 503/429 responses from Splunk Cloud             |
| Offset recovery        | Resetting offsets if the collector is offline > 12 hours       |

Source: Akamai TechDocs -- SIEM Integration Setup Guide

([techdocs.akamai.com/siem-integration/docs/akamai-siem-integration-for-splunk-and-cef-syslog](https://techdocs.akamai.com/siem-integration/docs/akamai-siem-integration-for-splunk-and-cef-syslog))

## 5. Splunk Components Involved

Multiple Splunk Cloud components participate in the Akamai integration. Each has a specific role and a specific installation location. Understanding the boundary between ingest-time and search-time processing is critical for troubleshooting.

| Component                         | What It Does                                                                                                                                                                                            | Where It Lives                                             | When It Acts                     |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------------------------------|
| <b>HEC (HTTP Event Collector)</b> | Receives JSON events from the external collector via HTTPS POST. Assigns index, sourcetype, and source based on token configuration.                                                                    | Splunk Cloud indexer tier (managed by Splunk)              | Ingest time                      |
| <b>Index</b>                      | Stores events. Created via Splunk Cloud UI (Settings > Indexes). Example: waf_akamai. Retention and access controls are configured here.                                                                | Splunk Cloud indexer tier                                  | Ingest time                      |
| <b>Sourcetype</b>                 | A label assigned to events (e.g., akamai:siem). Determines which parsing rules apply at search time. Set via HEC token metadata or collector configuration.                                             | Assigned at ingest, consumed at search time                | Both                             |
| <b>Akamai TA (App 4310)</b>       | The Akamai SIEM Integration add-on from Splunkbase. Provides JSON field extraction (KV_MODE=json), FIELDALIAS definitions, EVAL transformations, eventtypes, and CIM tags. Does NOT perform collection. | Splunk Cloud Search Head (installed via SSAI)              | Search time                      |
| <b>CIM Add-on</b>                 | Splunk Common Information Model (Splunkbase App 1621). Provides data model definitions, CIM Setup page, and validation datasets.                                                                        | Splunk Cloud Search Head (typically pre-installed with ES) | Search time                      |
| <b>Enterprise Security</b>        | The SIEM application. Consumes CIM-normalized data via accelerated data models. Provides correlation searches, notable events, dashboards, incident review, and threat intelligence framework.          | Splunk Cloud Search Head (ES-enabled stack)                | Search time + scheduled searches |

## Ingest-Time vs Search-Time Processing

| Phase              | What Happens                                                                                                                                                                       | Who Controls It                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Ingest Time</b> | Timestamp extraction, index assignment, sourcetype assignment, line breaking. These are set by HEC token metadata and any index-time transforms (typically minimal for JSON data). | Platform Engineering (HEC token + index configuration) |



| Phase       | What Happens                                                                                                                                                                                                          | Who Controls It                                          |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Search Time | JSON field extraction (KV_MODE=json), FIELDALIAS (vendor field to CIM field), EVAL (field transformations/normalizations), LOOKUP (value normalization), eventtype assignment, tag assignment, data model population. | Detection Engineering (TA installation + custom CIM app) |

**Critical Distinction:** The Akamai TA is installed on the search head for parsing and CIM mapping ONLY. It is not used for data collection. The TA's modular input capability is not used in this architecture. Collection is performed exclusively by the external SIEM connector.

## 6. CIM Mapping: What Akamai Provides vs What We Validate

### What Is CIM?

The Common Information Model (CIM) is a framework of standardized field names maintained by Splunk. It allows Enterprise Security to query data from different vendors using the same field names. For example, the attacker's IP address might be stored as **attackData.clientIP** in raw Akamai JSON, but ES looks for a field called **src**. CIM mapping creates the bridge between vendor-specific and standardized field names.

Without CIM mapping, ES cannot find events. Its dashboards, correlation searches, and threat intelligence lookups all query CIM field names -- not vendor field names. CIM mapping happens at search time via **FIELDALIAS**, **EVAL**, and **LOOKUP** definitions in the TA or a custom app.

### What the Akamai TA Provides

The Akamai SIEM Integration add-on (Splunkbase App ID 4310) provides baseline CIM coverage when installed on the search head. Per the add-on's configuration files, it provides:

| Capability            | Detail                                                                                                                    |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| JSON Field Extraction | KV_MODE=json for the akamai:siem sourcetype. Extracts nested JSON fields from attackData, httpMessage, and geo sections.  |
| CIM Field Aliases     | FIELDALIAS definitions that map vendor fields to CIM field names (e.g., attackData.clientIP to src).                      |
| Eventtypes and Tags   | Defines eventtypes (e.g., akamai_siem_attack) and assigns CIM-relevant tags (e.g., web, attack) for data model inclusion. |
| Lookup Definitions    | Lookup tables for normalizing Akamai-specific values to CIM-expected values (e.g., rule action normalization).            |

### Typical Vendor-to-CIM Field Mapping

The following table shows the most common mappings from Akamai vendor fields to Splunk CIM fields for the Web data model. This mapping must be validated -- the TA may not cover all fields, and field names may vary between Akamai product versions.

| Akamai Vendor Field | CIM Field   | CIM Data Model | Notes                                            |
|---------------------|-------------|----------------|--------------------------------------------------|
| attackData.clientIP | src         | Web            | Primary source IP of the attacker                |
| httpMessage.host    | dest        | Web            | Target hostname                                  |
| httpMessage.path    | url         | Web            | Request path (may need concatenation with query) |
| httpMessage.method  | http_method | Web            | GET, POST, HEAD, etc.                            |
| httpMessage.status  | status      | Web            | HTTP response status code                        |

| Akamai Vendor Field                       | CIM Field                   | CIM Data Model | Notes                                                             |
|-------------------------------------------|-----------------------------|----------------|-------------------------------------------------------------------|
| attackData.ruleActions                    | action                      | Web            | Requires normalization: deny to blocked, alert/monitor to allowed |
| httpMessage.requestHeaders.<br>User-Agent | http_user_agent             | Web            | May require extraction from nested headers                        |
| httpMessage.bytes                         | bytes                       | Web            | Response body size                                                |
| geo.country                               | src_country                 | Web            | ISO country code of attacker                                      |
| attackData.configId                       | vendor_product<br>(partial) | Web            | Often requires EVAL to set vendor_product='Akamai WAF'            |

## Why CIM Is Never "Done" by Default

- The TA provides **partial** CIM coverage. Some fields (like vendor\_product, app) are often not mapped by the vendor TA and require manual EVAL definitions.
- CIM field population depends on the Akamai vendor fields actually being present in the event. If a field is empty in the raw JSON, the CIM alias produces nothing.
- The **action** field requires normalization from Akamai values (deny, alert, monitor) to CIM-expected values (blocked, allowed). This is a common gap.
- Tag assignments (web, attack) must match the data model's constraint definitions. If tags are missing, events will not appear in the data model even if CIM fields are populated.
- Different Akamai products (WAF vs Bot Manager) may have different field schemas. A single set of FIELDALIAS rules may not cover all event types.

**Fact-Check Statement:** Manual CIM extension is a normal and expected part of Akamai onboarding. The TA provides a baseline; the engineering team must validate and extend it. This is consistent with Splunk's CIM documentation, which states that add-on developers make 'best effort attempts' to map recommended fields.

## 7. Step-by-Step CIM Validation Workflow

**Environment:** All validation steps below must be performed in the **Dev Splunk Cloud environment** first. Production is never used for validation or experimentation.

### Step 7.1 -- Inspect Raw Events

Before validating CIM, confirm that raw events are arriving and that vendor fields are extractable.

```
index=<dev_index> sourcetype=<sourcetype> earliest=-1h
| head 10
| table _time, _raw
```

Verify: Events contain JSON with attackData, httpMessage, and geo sections. Timestamps are correct (not in the future, not days old).

### Step 7.2 -- Check Vendor Field Population

```
index=<dev_index> sourcetype=<sourcetype> earliest=-1h
| fieldsummary
| search field IN ("attackData.clientIP", "attackData.configId",
"httpMessage.host", "httpMessage.method", "httpMessage.status",
"httpMessage.path", "attackData.ruleActions", "geo.country")
| table field, count, distinct_count, is_exact, numeric_count
```

**Success:** All fields show count > 0 with reasonable distinct\_count values. **Failure:** Fields show count = 0, indicating the TA is not installed, the sourcetype does not match, or the JSON structure has changed.

### Step 7.3 -- Check CIM Field Population

```
index=<dev_index> sourcetype=<sourcetype> earliest=-1h
| fieldsummary
| search field IN ("src", "dest", "url", "http_method",
"status", "action", "http_user_agent", "vendor_product", "app")
| table field, count, distinct_count
```

**Success:** Core CIM fields (src, dest, action, http\_method) show count > 0 with > 90% population relative to total event count. **Failure:** CIM fields show count = 0 or very low population. This indicates missing FIELDALIAS or EVAL definitions -- the TA may need to be supplemented with a custom CIM app.

### Step 7.4 -- Validate CIM Tags

```
index=<dev_index> sourcetype=<sourcetype> earliest=-1h
| head 1
| table tag::*
```

Verify that tags include **web** (for the Web data model) and optionally **attack** (for Intrusion\_Detection). Without these tags, events will not appear in the target data model regardless of CIM field population.

### Step 7.5 -- Validate Web Data Model Population

```
| tstats count FROM datamodel=Web
WHERE (index=<dev_index> sourcetype=<sourcetype>)
BY sourcetype
```

**Success:** Returns a row with count > 0 for your sourcetype. **Failure:** Returns no results. Check: (1) CIM tags are assigned, (2) the index is in the data model's allowed index list (CIM Setup page), (3) acceleration is enabled and building.

## Step 7.6 -- Inspect CIM Field Values in the Data Model

```
| tstats count FROM datamodel=Web
WHERE (index=<dev_index> sourcetype=<sourcetype>)
BY Web.src, Web.dest, Web.url, Web.http_method, Web.status, Web.action
| head 20
```

**Success:** CIM fields show meaningful, non-null values. **Failure:** Fields show null or empty -- CIM mapping exists but the underlying vendor field is empty or the alias is incorrect.

## Step 7.7 -- Remediation: Create Custom CIM App

If CIM validation reveals gaps, create a custom CIM app. **Never edit the vendor TA directly** -- vendor updates will overwrite your changes.

Create a separate app (e.g., SA-akamai-cim-overrides) with a props.conf that adds missing FIELDALIAS and EVAL definitions for the Akamai sourcetype. Deploy this app via SSAI to the Dev search head first. After validation passes, promote the identical app to Prod.

```
# Example props.conf for SA-akamai-cim-overrides
[akamai:siem]
FIELDALIAS-akamai_src = attackData.clientIP AS src
FIELDALIAS-akamai_dest = httpMessage.host AS dest
FIELDALIAS-akamai_url = httpMessage.path AS url
FIELDALIAS-akamai_method = httpMessage.method AS http_method
FIELDALIAS-akamai_status = httpMessage.status AS status
EVAL-action = case(
match(attackData.ruleActions, "(?i)deny"), "blocked",
match(attackData.ruleActions, "(?i>alert"), "allowed",
match(attackData.ruleActions, "(?i)monitor"), "allowed",
true(), attackData.ruleActions)
EVAL-vendor_product = "Akamai WAF"
EVAL-app = "Akamai App and API Protector"
```

*Note: The exact vendor field names depend on the Akamai TA version and event schema. Always validate against actual events in your Dev environment.*

## 8. Dev to Prod Safety Model

**Mandatory Rule:** All Akamai WAF/DDoS changes must be validated in Dev before promotion to Prod. This prevents breaking Splunk Enterprise Security in production.

### Why Dev Testing Is Mandatory

Enterprise Security does not maintain separate CIM configurations per data source. CIM field aliases, EVAL rules, eventtypes, and tags operate at the **sourcetype level across the entire search head**. If a change is incorrect, it can affect every data source that shares the same CIM data model or field names -- not just Akamai.

### How CIM Changes Can Break ES

| Risk                               | What Happens                                                                                                                       | Prevention                                                                                      |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Incorrect FIELDALIAS               | Wrong values populate CIM fields. ES correlation searches produce false positives or false negatives across all data sources.      | Validate every alias in Dev with real data before deploying to Prod.                            |
| Overly broad tags                  | Non-WAF events get pulled into the Web data model, polluting it with events that lack required CIM fields.                         | Scope eventtypes narrowly. Test in Dev with   datamodel Web search   stats count by sourcetype. |
| Conflicting EVAL overrides         | Two TAs define EVAL-action for different sourcetypes but one has a global scope. Last-loaded conf wins, breaking the other source. | Always scope to a specific [sourcetype] stanza. Never use [default].                            |
| Data model acceleration corruption | Tag changes on an already-accelerated data model make summaries stale. ES uses old summaries until rebuild completes.              | After any tag change in Dev, rebuild acceleration and verify.                                   |
| Editing vendor TA directly         | Changes are silently overwritten on TA update via SSAI, creating a 'worked yesterday, broken today' failure.                       | Never edit vendor TA. All custom mappings go in a separate app.                                 |

### Dev vs Prod Responsibilities

| Activity              | Dev Environment                                    | Prod Environment                                   |
|-----------------------|----------------------------------------------------|----------------------------------------------------|
| Collector deployment  | Deploy and configure connector (Dev VM to Dev HEC) | Deploy clone with Prod HEC endpoint + token        |
| TA installation       | Install via SSAI, validate JSON extraction         | Install same version via SSAI after Dev validation |
| CIM custom app        | Create, iterate, test SA-akamai-cim-overrides      | Deploy identical tested version via SSAI           |
| CIM validation        | Run all fieldsummary + tstats checks               | Run identical checks to confirm parity             |
| Data model validation | Verify tstats returns, check acceleration          | Confirm data model population matches Dev          |

| Activity             | Dev Environment                               | Prod Environment                                   |
|----------------------|-----------------------------------------------|----------------------------------------------------|
| Detection enablement | Test correlation searches                     | Enable only after all Prod gates pass              |
| Experimentation      | Safe to iterate on field aliases, tags, evals | NO experimentation. Only promoted, tested configs. |

## The Safe Change Protocol

| Step | Action                                                           |
|------|------------------------------------------------------------------|
| 1    | Make the change in Dev only (custom CIM app, not vendor TA)      |
| 2    | Run CIM validation (fieldsummary + tstats checks)                |
| 3    | Check data model impact: verify only expected sourcetypes appear |
| 4    | Rebuild data model acceleration if tags changed                  |
| 5    | Run all acceptance gates in Dev                                  |
| 6    | Deploy identical change to Prod. Run gates again in Prod.        |

## 9. Detection Readiness (No RBA Assumption)

### ES Framework vs Correlation Logic

Enterprise Security provides a framework -- it does not auto-detect threats simply because data is ingested. ES includes pre-built correlation searches, but these must be explicitly enabled, tuned, and often customized for the environment. CIM compliance enables correlation but does not create it.

The ES framework consists of: (1) data models that organize CIM-normalized data, (2) correlation searches that query data models for patterns, (3) notable events generated when a correlation search matches, and (4) incident review workflows for analyst triage. Each layer depends on the one below it. If CIM is broken, the entire chain fails silently.

### What 'Detection Ready' Actually Means

| Criterion                          | How to Verify                                                              |
|------------------------------------|----------------------------------------------------------------------------|
| Events are flowing continuously    | SPL: index=...   timechart span=1h count. No zero-count hours.             |
| CIM fields are populated at > 90%  | SPL: fieldsummary filtered to CIM fields. Count matches total event count. |
| Target data model is populated     | SPL:   tstats count FROM datamodel=Web WHERE index=... returns > 0.        |
| Data model acceleration is healthy | Settings > Data Models > check completion percentage > 90%.                |
| Tags are correctly assigned        | SPL:   head 1   table tag::* shows 'web' and/or 'attack' tags.             |
| Events pass all acceptance gates   | Ingestion, lag, parsing, CIM, data model gates all show PASS.              |

### When Use Cases Should Be Written

Detection use cases (correlation searches) should only be written after all of the above criteria are met in both Dev and Prod. Writing correlation searches against broken or incomplete data creates false confidence -- detections appear to work but silently miss real threats.

- Wait at least 1-2 weeks of stable data flow before setting thresholds
- Do not enable RBA risk scoring -- the organization is not using RBA
- Do not build complex multi-source correlations until other data sources (endpoint, identity, network) are also CIM-mapped and validated
- Start with simple, high-confidence detections: blocked WAF events, excessive denied requests from a single source, known-bad User-Agent strings

**Key Distinction:** 'Data readiness' (this document) and 'detection readiness' (use case development) are two separate milestones. This document ends at data readiness. Detection use cases are a subsequent workstream.



## 10. Key Takeaways for Management

The following summarizes the critical points from this document in plain language, suitable for communication to security engineering management.

| Topic                                 | Summary                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Why Akamai onboarding is non-trivial  | Akamai security events are not pushed to Splunk. An external collector must be deployed, configured, and maintained on our infrastructure. The collector polls the Akamai API, manages authentication, handles offsets, and forwards data to Splunk Cloud. This is infrastructure we own and operate -- Akamai does not manage it, and Splunk Cloud cannot run it.            |
| Why CIM validation matters            | Enterprise Security does not read raw vendor fields. It relies on standardized CIM field names (src, dest, action, http_method). The Akamai add-on provides partial CIM coverage, but gaps are expected and must be closed with a custom app. Without CIM validation, ES dashboards, correlation searches, and threat intelligence lookups will not find Akamai events.       |
| Why collector ownership is required   | The collector is the only component that holds Akamai API credentials and communicates with Akamai infrastructure. If the collector is offline for more than 12 hours, events are permanently lost (Akamai retains events for 12 hours only). Collector monitoring, patching, and credential rotation are ongoing operational responsibilities.                               |
| Why this approach is safe and correct | All changes are validated in a Dev Splunk Cloud environment before touching Prod. CIM changes are scoped to specific sourcetypes using a separate custom app -- the vendor TA is never modified. Data model acceleration is verified before enabling detections. This prevents the risk of breaking ES in production, which would affect all data sources -- not just Akamai. |
| Why multiple config IDs matter        | Enterprise Akamai deployments typically have multiple SIEM config IDs -- one per security product, web property, or business unit. The collector must be configured with all relevant config IDs. Index strategy (shared vs separate indexes) must be decided based on access control and retention requirements. This adds operational complexity that must be planned for.  |

## Official Documentation References

| Source                      | URL                                                                                                                                                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Akamai SIEM Integration API | <a href="https://techdocs.akamai.com/siem-integration/reference/api">techdocs.akamai.com/siem-integration/reference/api</a>                                                                                   |
| Akamai SIEM Setup Guide     | <a href="https://techdocs.akamai.com/siem-integration/docs/akamai-siem-integration-for-splunk-and-cef-syslog">techdocs.akamai.com/siem-integration/docs/akamai-siem-integration-for-splunk-and-cef-syslog</a> |
| Akamai SIEM Response Format | <a href="https://techdocs.akamai.com/siem-integration/reference/multi-json-response-format-1">techdocs.akamai.com/siem-integration/reference/multi-json-response-format-1</a>                                 |
| Akamai ULS (GitHub)         | <a href="https://github.com/akamai/ULS">github.com/akamai/ULS</a>                                                                                                                                             |
| Splunk CIM Documentation    | <a href="https://docs.splunk.com/Documentation/CIM">docs.splunk.com/Documentation/CIM</a>                                                                                                                     |
| Splunk Web Data Model       | <a href="https://docs.splunk.com/Documentation/CIM/latest/User/Web">docs.splunk.com/Documentation/CIM/latest/User/Web</a>                                                                                     |

| Source                         | URL                                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------|
| Akamai TA (Splunkbase)         | <a href="https://splunkbase.splunk.com/app/4310">splunkbase.splunk.com/app/4310</a> |
| Splunk CIM Add-on (Splunkbase) | <a href="https://splunkbase.splunk.com/app/1621">splunkbase.splunk.com/app/1621</a> |

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